have been snapped up and married, and all the nice problems have been solved. I don't really like any of those that are left," he said in a moment of despair in Berlin in 1929. But, in fact, explorations of the *consequences* of the laws of quantum mechanics and relativity were only beginning, and those consequences would hold wonderful surprises: the structure of the atomic nucleus, nuclear energy, black holes and their evaporation, superfluidity, superconductivity, transistors, lasers, and magnetic resonance imaging, to name only a few. And Landau, despite his pessimism, would become a central figure in the quest to discover these consequences.

Upon his return to Leningrad in 1931, Landau, who was an ardent Marxist and patriot, resolved to focus his career on transfusing modern theoretical physics into the Soviet Union. He succeeded enormously, as we shall see in later chapters.

Soon after Landau's return, Stalin's iron curtain descended, making further travel to the West almost impossible. As George Gamow, a Leningrad classmate of Landau's, later recalled: "Russian science now had become one of the weapons for fighting the capitalistic world. Just as Hitler was dividing science and the arts into Jewish and Aryan camps, Stalin created the notion of capitalistic and proletarian science. It [was becoming] ¼ a crime for Russian scientists to 'fraternize' with scientists of the capitalistic countries."

The political climate went from bad to horrid. In 1936 Stalin, having already killed 6 or 7 million peasants and kulaks (landowners) in his forced collectivization of agriculture, embarked on a several-year-long purge of the country's political and intellectual leadership, a purge now called the Great Terror. The purge included execution of almost all members of Lenin's original Politburo, and execution or forced disappearance, never to be seen again, of the top commanders of the Soviet army, fifty out of seventy-one members of the Central Committee of the Communist party, most of the ambassadors to foreign countries, and the prime ministers and chief officials of the non-Russian Soviet Republics. At lower levels roughly 7 million people were arrested and imprisoned and 2.5 million died—half of them intellectuals, including a large number of scientists and some entire research teams. Soviet biology, genetics, and agricultural sciences were destroyed.

In late 1937 Landau, by now a leader of theoretical physics research in Moscow, felt the heat of the purge nearing himself. In panic he searched for